

Modeling Cross-Linguistic Perfect-Like Grams Through Parallel Corpus Data

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Project Overview

What even is a 'Perfect'?

If you ask three linguists what a 'Perfect' is, you'll get four different answers, and that's just in English.

- Properties of both Tense + Aspect
- “Dual temporal layer”
- Anteriority and “current relevance”

! Various recent studies^a have called into question whether 'Perfect' is even a viable category for cross-linguistic comparison and analysis.

^ae.g., Bertrand et al. (2022), Bowler & Ozkan (2017), Zhao (2022a, b)

What even is a 'Perfect'?

This project uses the term 'Perfect-like gram' ('PLG') where

- a *gram* refers to a grammatical marker (such as an affix, particle, or auxiliary) that encodes a recurrent meaning or function within a language, and
- 'Perfect-like' refers to grams that encode at least one of the three canonical readings of the Perfect: Resultative, Experiential, Universal.

This project introduces a novel methodology: applying an ontological feature schema to train NLP classifiers (Random Forest and MLP) on parallel corpus data, improving model predictiveness.



Features include, for example, specific readings, Aktionsart/'lexical aspect', and modality distinctions.

Methodology (Pilot Study)

This project used verse-aligned parallel New Testament Bible data in English, Mandarin, and German. Once the data were aligned, all verses that in English include auxiliary *have* + participle were identified and extracted using a variety of Python-based toolkits.

First, each verse in the extracted dataset was scanned to identify the PLG in its text.

Next, the verse was annotated for feature values based on the English text.

Finally, the parallel texts were analyzed to determine how the PLG in the English text is expressed in each of the other languages.

Running the Experiments

The annotated data were exported in CSV format and submitted to both classifiers. The data was structured such that each row represents a unique occurrence of a PLG (each with its own unique token ID), with columns representing binary feature values and language-specific grams.



Non-English tokens that were deemed non-equivalent were marked and filtered from the dataset prior to training.

Data Structure

TokenID	target	Exp	Res	Univ	DTAM	Evid	TAML	NegPol	akt_activity	akt_accomp	akt_achiev	akt_state	akt_semel	eng_constr	deu_constr	cmn_constr
40001018	had been promised	0	1	0	0	0	1	0	0	0	1	0	0	pastperf	pastperf	le
40001024	had commanded	1	0	0	0	0	1	0	0	0	1	0	0	pastperf	pastperf	suo_de
40003002	has come near	0	1	0	0	0	0	0	0	1	0	0	0	perf	perf	le
40003017	have approved	0	1	0	0	0	0	0	0	0	1	0	0	perf	perf	suo_de
40004012	had been arrested	0	1	0	0	1	1	0	0	0	1	0	0	pastperf	pastperf	unmarked
40005027	have heard	1	0	0	0	0	0	0	0	0	1	0	0	perf	perf	guo
40005038	have heard	1	0	0	0	0	0	0	0	0	1	0	0	perf	perf	guo
40005043	have heard	1	0	0	0	0	0	0	0	0	0	0	0	perf	perf	guo
40006002	have been paid	0	1	0	0	0	0	0	0	0	1	0	0	perf	n_a	yijing_le
40006005	have been paid	0	0	0	0	0	0	0	0	0	1	0	0	perf	n_a	yijing_le
40006016	they have received	0	1	0	0	0	0	0	0	0	1	0	0	perf	n_a	yijing_le
40007028	had finished	0	1	0	0	0	0	0	0	1	1	0	0	pastperf	pastperf	le
40008010	have not found	1	0	0	0	0	0	1	0	0	1	0	0	perf	perf	meiyou_guo
40008029	have you come	0	1	0	0	0	0	0	0	1	0	0	0	perf	perf	n_a
40009008	had given	0	1	0	0	0	1	0	0	0	1	0	0	pastperf	perf	n_a
40009022	has made you well	0	1	0	0	0	0	0	0	1	0	0	0	perf	perf	le
40009033	have never seen	1	0	0	0	0	0	1	0	0	1	0	0	perf	perf	meiyou
40010023	have gone through	0	1	0	0	0	0	0	0	1	0	0	0	perf	n_a	meiyou

Once the models were trained and tested, the results for each language-classifier pair were exported to a confusion matrix, as well as to itemized error reports for analysis using a variety of statistical performance metrics.

Results (Pilot Study)

Big Picture Takeaway

Across languages and classifiers, the dominant pattern was not diffuse errors but clustering: a small set of feature contrasts disproportionately accounted for prediction failures.

Both classifiers performed best in the presence of the Resultative feature, and worst in the presence of the Experiential feature, and results for both features tested as significant according to χ^2 scoring (i.e., p -value < 0.05).



This suggests that the models interpret the Mandarin TAM system as organized primarily around a +/- Resultative distinction.

One particularly common error type feature association for both models was the overprediction of *le* for actual *suo...de* and nullmarked grams.

- All instances of this error type were +Resultative
- Actual *suo...de* were primarily Accomplishments
- Actual nullmarked were primarily Achievements



This further suggests an Aktionsart feature association, but more data are needed to posit solid conclusions.

The German results include two salient patterns:

1. Both models struggled with the TAML¹ feature
2. Both models treat universals as a salient category, but Aktionsart interacts with them in misleading ways

¹TAM Layering

Both models overpredicted the German Plusquamperfekt² for all +TAML tokens, many of which were actual modals and Konjunktiv II mood³.



TAML is a coarse-grained feature which includes any additional tense, aspect, and/or mood component in a PLG. This feature has been decomposed into several sub-types for the next set of experiments.

²Past Perfect/Pluperfect

³Analogous to the Subjunctive

Both models consistently confused the Perfekt (Perfect) and the Präsens (Present) in +Universal tokens in a way that split along Aktionsart lines:

- **Universal Activities** were predicted to be Perfekt but actual Präsens
- **Universal States** were predicted to be Präsens but actual Perfekt

This indicates that the models' handling of universals is not random but conditioned by predicate class.



Why this should be is a matter that is currently under consideration.

Both models handled English extremely well, perhaps not surprisingly considering that English was used as the anchor text for feature and gram valuation.

The one significant error type that occurred in both classification experiments involved TAM Layering, echoing the German TAML errors.

Continuing Work

Several methodological refinements have been made pursuant to the continuing work, most notably:

- Feature decomposition, especially of TAM Layering
- Gram decomposition for more granular analysis of paradigm oppositions
- L1 annotators have been recruited to assist with identifying grams in non-English text as well as differences in meaning or nuance between the English and non-English texts

Thank you very much!
Questions are welcome 😊